

Neuroscience

Octopamine integrates the status of internal energy supply into the formation of food-related memories

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint posted

July 3, 2023 (this version)

Sent for peer review

May 15, 2023

Posted to bioRxiv

May 5, 2023

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Abstract

Summary

The brain regulates food intake in response to internal energy demands and the availability of food. However, can internal energy storage influence the kind of memory that is formed? We show that the duration of starvation determines whether *Drosophila melanogaster* forms appetitive short-term or long-term memory. The internal glycogen storage in the muscles and fat tissue influences how long sucrose is remembered. Insulin-like signaling in octopaminergic reward neurons integrates internal energy storage into memory formation. In turn, octopamine suppresses the formation of long-term memory. Octopamine is not required for short-term memory, since octopamine-deficient mutants form appetitive short-term memory to sucrose and to other nutrients depending on the internal energy status. The reduced positive reinforcing effect of sucrose in the presence of high internal glycogen levels combined with the increased stability of food-related memories due to increased starvation periods might drive increased food intake.

eLife assessment

This **important** study dissects the interplay between internal energy homeostasis, food intake and food-related memories, focusing on the role played in *Drosophila* by Octopamine, a monoamine functionally related to noradrenalin. This work will help interpret previously published work on the function of Octopamine, and should be of interest to the growing community of biologists interested in how internal state shapes behavior, including decision making processes, learning and memory. The experimental evidence is **solid**, but additional evidence using, for instance, tissue-specific and effector-specific manipulation, and a unifying model would strengthen the present study.

Introduction

The internal energy status of an animal needs to be adjusted to the energy expenditure of the organism and the availability of external food to ensure survival. Increased storage of energy correlates with increased food intake in the past. Dysregulation of food intake might result in diseases such as obesity and diabetes. Sucrose is a carbohydrate enriched in Western diets, and the breakdown product of sucrose - glucose - can be stored in the organism as glycogen mainly in the liver and muscles. Increased glycogen levels are a hallmark of glycogen storage diseases that are accompanied by defects in the liver, muscles and brain.([Ellingwood and Cheng, 2018](#)) Similar to vertebrates, the fruit fly *Drosophila melanogaster* uses glucose as a primary energy source and stores glycogen mainly in the muscles—the major site of energy expenditure—and the fat body—the equivalent of the vertebrate liver.([Galikova and Klepsatel, 2023](#); [Wigglesworth, 1949](#)) As in vertebrates, glycogen levels are also found in the brain.([Yamada et al., 2018](#))

The peptide hormone insulin regulates glucose homeostasis at the cellular level.([Saltiel and Kahn, 2001](#)) Insulin and its *Drosophila melanogaster* counterpart, insulin-like peptides, perform their functions through well-conserved signal transduction cascades.([Chatterjee and Perrimon, 2021](#); [Inoue et al., 2018](#)) In vertebrates, in addition to its function in fat tissue and muscles, the insulin receptor is broadly expressed in the brain and regulates neuronal plasticity.([Nakai et al., 2022](#)) For example, in rats, reduced insulin receptor function in the hypothalamus results in loss of long-term potentiation and impaired spatial memory.([Grillo et al., 2015](#)) In *Drosophila melanogaster*, the insulin receptor substrate Chico is required for the development of mushroom bodies, a brain region required for learning and memory, and loss of Chico results in learning defects.([Naganos et al., 2012](#)) Disrupting the function of the insulin receptor in the mushroom bodies also results in learning defects, whereas the insulin receptor and Chico are required in the ellipsoid body for protein biosynthesis-dependent long-term memory.([Chambers et al., 2015](#)) While there is evidence that insulin receptor signaling regulates cellular physiology, it is not clear how the internal energy supply of the animal influences the formation of food-related memories.

In *Drosophila melanogaster*, the monoamine octopamine - functionally related to noradrenalin in vertebrates - is involved in the regulation of energy homeostasis. Upon starvation, the glucose and trehalose concentrations in the hemolymph change less in tyramine- β -hydroxylase (*Tbh*) mutants lacking the neurotransmitter octopamine, and in addition, the life span of *Tbh* mutants is extended.([Damrau et al., 2017](#); [Li et al., 2016](#)) *Tbh* mutants show an increased threshold to respond to sucrose with the extension of their proboscis when sucrose is offered to their tarsi – a structure that contains gustatory receptor neurons. The reduced response threshold to sucrose correlates with a reduced sucrose intake in *Tbh* mutants.([Li et al., 2016](#); [Scheiner et al., 2014](#)) Octopaminergic neurons potentiate the response of sugar-sensing gustatory receptor neurons in satiated flies, suggesting that octopaminergic neurons regulate feeding behavior by changing the sensitivity of taste receptor neurons.([Youn et al., 2018](#)) The reduced responsiveness to sucrose is also thought to be responsible for defects observed in habituation in *Tbh* mutants.([Scheiner et al., 2014](#)) In addition to their changes in simple forms of neuronal plasticity, *Tbh* mutants fail to form a positive association with a food reward in a classical olfactory conditioning paradigm. Flies quickly learn to associate an odorant with sucrose, and *Tbh* mutants do not show this positive association directly after learning.([Schwaerzel et al., 2003](#)) However, when a longer time after training has elapsed, memory appears.([Das et al., 2014](#)) The release of octopamine mediates the reinforcing effect of sweet taste in short-term memory and the anesthesia-resistant form of long-term memory.([Burke et al., 2012](#); [Huetteroth et al., 2015](#); [Wu et al., 2013](#))

To analyze the relationship between energy status, the evaluation of a food reward and the formation of memory and food intake, we used the olfactory associative learning and memory paradigm. Using nutrients as a positive reward allows for investigating how the internal energy status influences the evaluation of the reinforcer and in turn learning and memory. As a step to understand the interconnection between reward evaluation, nutrient intake and the formation of food-related memories, we analyzed the function of octopamine in the regulation of internal energy homeostasis, learning and memory and food intake.

Results

Starvation increases learning and memory performance when using a food-related reward (Colomb et al., 2009). However, how the internal energy status is integrated into the evaluation of the reward and the stability of food-related memories and whether there is a correlation between the perceptions of the food reward, food-related memories and food intake are not clear.

The duration of starvation influences the strength and stability of memories

To investigate whether changes in internal energy storage influence food-related memories, we used the olfactory associative learning and memory paradigm using sucrose as a food-related reinforcer. (Schwaerzel et al., 2003) In the paradigm, flies learn to associate an odorant with sucrose and are later tested on whether they remember the association. Starved flies were exposed for 2 min to an unconditioned odorant, 4-methylcyclohexanol (MCH), followed by a 2 min exposure to a second odorant, 3-octanol (3-OCT), that was paired with the reward of 2 M sucrose. After the training, flies were given a choice between the two odorants (Figure 1A). Normally, flies have formed a positive association between the reward and the odorant within 2 min after the training. In a reciprocal experiment, the second odorant was rewarded. The learning index presents the average positive association between the reward and both odorants. Before the experiments, flies were tested to determine whether they perceived the odorants, preferred one odorant over other and responded to the reward similarly to ensure that the observed differences in behavior were not due to changes in odorant perception or sucrose sensitivity (Table S1). To alter the internal energy status, flies were starved either for 16 h or for 40 h before the training (Figure 1). We used 3- to 5-day-old male flies to minimize differences in body weight and to control for differences in food preferences. We included $T\beta h^{nM18}$ mutants lacking octopamine in the experiments, since they showed defects in energy metabolism and sucrose reward learning. (Li et al., 2016; Schwaerzel et al., 2003) Flies that were starved for 16 h formed a positive association between the rewarded odorants and remembered this association for 2 min but not for 6 h. Consistent with previous results, learning and memory performance was significantly improved by prolonged starvation. (Colomb et al., 2009) In addition, 16 h starved $T\beta h^{nM18}$ mutants did not show 2 min memory but emerged memory 6 h after training. (Das et al., 2014; Schwaerzel et al., 2003) In contrast, after 40 h of starvation, $T\beta h^{nM18}$ mutants already remembered the sucrose paired odorant 2 min after training to a similar extent as after 6 h (Figure 2B). We repeated similar experiments using a lower concentration of 0.15 M sucrose as a reinforcer (Figure S1). Here, prolonged starvation significantly increased learning and memory performance in controls and $T\beta h^{nM18}$ mutants. In contrast to the experiments using 2 M sucrose as a reward, 16 h starved $T\beta h^{nM18}$ mutants form a negative association with 0.15 M sucrose 2 min after training, but when starved for

40 h, form a positive 2 min memory. Starvation and reinforcer strength regulate appetitive memory strength.

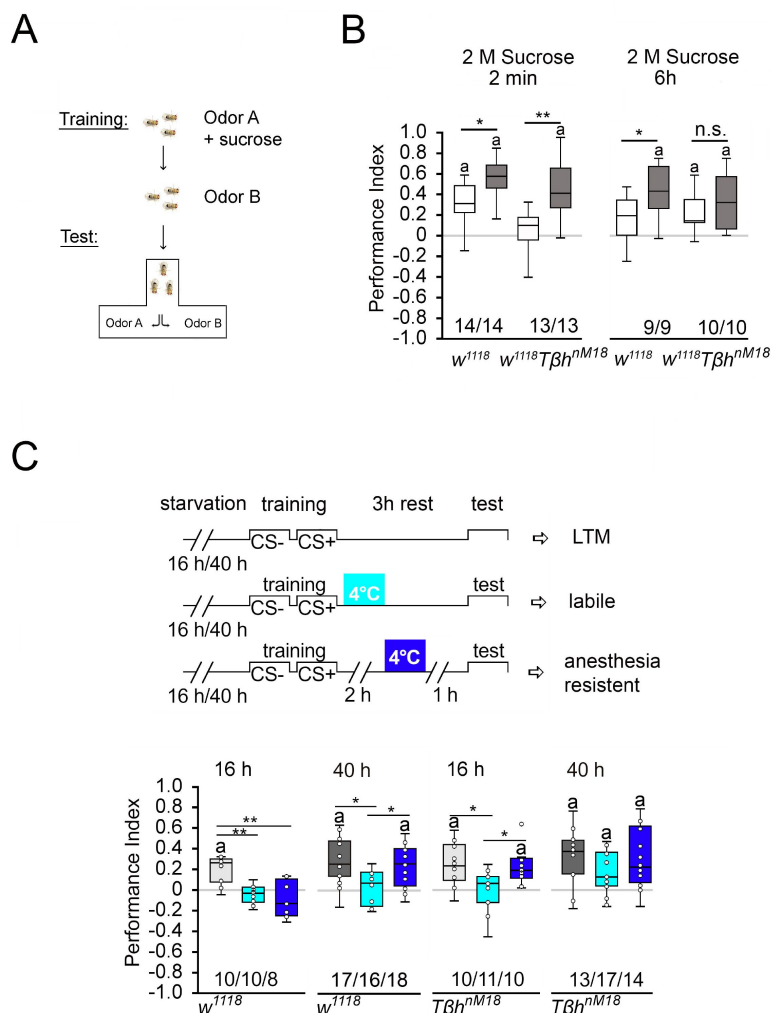


Figure 1.

Starvation influences the strength and stability of memory

A, Appetitive olfactory learning and memory paradigm. **B**, Appetitive 2 min STM and 6 h memory in w^{1118} and $T\beta h^{nM18}$ after 16 h (white bars) or 40 h starvation (dark gray bars) before the training using 2 M sucrose as reinforcer. Prolonged starvation increases memory performance. Prolonged starvation of $T\beta h^{nM18}$ for 40 h resulted in STM.

Independent of the duration of starvation, 6 h after training, memory appeared in the mutants. **E**, Appetitive STM training with 2 M sucrose and cold shock directly or 2 h after training. Prolonged starvation shifts in control STM to LTM and in $T\beta h^{nM18}$ LTM to ARM. Numbers below box blots indicate pairs of reciprocally trained independent groups of male flies. Student's t tests were used to determine differences between two groups, and one-way ANOVA with Tukey's post hoc HSD test was used to determine differences between three groups. The letter "a" marks a significant difference from random choice as determined by a one-sample sign test ($P^* < 0.05$; $P^{**} < 0.01$).

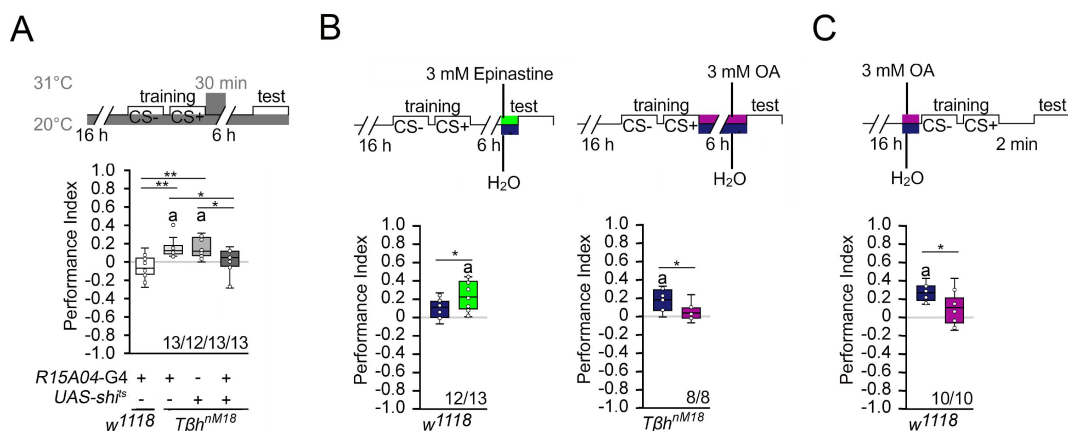


Figure 2.

Octopamine acts upstream of dopamine and negatively regulates memory

A, A 30 min block of dopaminergic PAM neurons required for LTM after training using a temperature-sensitive *shibire* transgene under the control of the R1504-Gal4 driver results in *Tβh^{NM18}* mutants losing LTM. **B**, Feeding 3 mM of the octopamine receptor antagonist Epinastine for 1 h directly after training resulted in *w¹¹¹⁸* flies in memory 6 h later. Feeding 3 mM octopamine for 6 h after training suppresses LTM in *Tβh^{NM18}*. **C**, A 3 mM octopamine feeding pulse 30 min before training inhibits STM in *Tβh^{NM18}* mutants. Controls were water-fed. One-way ANOVA with Tukey's HSD post hoc test was used to determine differences between three groups, and Student's t tests were used for two groups. The letter "a" marks a significant difference from random choice as determined by a one-sample sign test ($P^* < 0.05$; $P^{**} < 0.01$).

Next, we analyzed how starvation influences the stability of memory. Protein synthesis-dependent long-term memory is still labile directly after training and can be blocked by 4°C cold-shock anesthesia directly after training (Krashes and Waddell, 2008). After consolidation, long-term memory is cold-shock resistant and insensitive to a cold shock 1 h before the test. In addition, a second form of longer lasting memory exists, which is directly after training cold-shock insensitive (Figure 1C). To investigate how starvation influences the stability of appetitive memory, we applied cold-shock anesthesia at different time points after training and analyzed the influence on memory formation in control and *Tβh^{NM18}* mutant flies (Figure 1C). As expected 3 h after training, memory was still present in 16 h starved flies and was completely abolished by a cold shock directly after training or shortly before the test. Increased starvation resulted in memory that was sensitive to cold shock directly after training but not shortly before the test. Thus, an extended period of starvation resulted in long-term memory in control flies. In contrast to 16 h starved controls, 16 h starved *Tβh^{NM18}* mutants developed long-term memory that was sensitive to cold shock directly after training and cold shock insensitive shortly before the test. Longer periods of starvation resulted in *Tβh^{NM18}* mutants in anesthesia-resistant memory. Thus, with increasing length of starvation, memory becomes more stable. Depending on the duration of starvation, animals first form STM memory, then cold-shock-sensitive LTM and later ARM. The *Tβh^{NM18}* mutants that were starved for 16 h formed similar memories to the 40 h starved control flies.

Octopamine is a negative regulator of memory

Starvation induces the formation of protein synthesis-dependent long-term memory. (Krashes and Waddell, 2008) Thus, it is likely that the emerging memory 6 h after training in $T\beta h^{nM18}$ mutants is long-term memory. To address this, we abolished a mechanism specifically required for long-term memory in $T\beta h^{nM18}$ mutants and analyzed whether this interferes with emerging memory (Figure 2). The R15A04-Gal4 driver targets dopaminergic neurons of the PAM cluster specifically required for appetitive long-term memory but not short-term memory. (Yamagata et al., 2015) Blocking the function of these dopaminergic neurons directly after training using a temperature-sensitive *shibire* transgene (UAS-shi^{ts}) and a 30 min long heat pulse of 31°C resulted in loss of memory in $T\beta h^{nM18}$ mutants (Figure 2A). This is consistent with the fact that emerging memory is long-term memory. Since the $T\beta h^{nM18}$ mutants lack the neurotransmitter octopamine, we next investigated whether octopamine normally suppresses LTM. To address this, we first blocked the function of octopamine receptors in controls directly after the training by feeding the octopamine antagonist epinastin for 1 h and analyzed memory 5 h later (Figure 2B). If the function of the octopamine receptor is required for long-term memory, longer lasting memory should also appear in control flies that are starved for 16 h prior to the training, which indeed was the case. Consistent with the idea that octopamine is a negative regulator of long-term memory, feeding octopamine to $T\beta h^{nM18}$ mutants directly after training blocked long-term memory (Figure 2B). To analyze whether octopamine is also able to block STM, we fed octopamine prior to training to control flies (Figure 2C). A short pulse of octopamine before the training inhibits the STM. Thus, octopamine is a negative regulator of appetitive dopaminergic neuron-dependent long-term memory and can block STM.

Starvation influences sucrose consumption preference

Starvation reduces internal energy storage. The reduction might result in reevaluation of external food cues and increased food consumption to restore the energy supply. To investigate whether starvation depletes glycogen storage, we measured glycogen levels in whole animals (Figure 3A). In controls and $T\beta h^{nM18}$ mutants, starvation reduces glycogen levels; however, non-starved $T\beta h^{nM18}$ mutant males started out with significantly higher glycogen levels. After 40 h of starvation, the glycogen levels of $T\beta h^{nM18}$ mutants were still higher than those of controls that were starved for the same amount of time. The reevaluation of external food cues might be reflected in the choice of food. Adult $T\beta h^{nM18}$ mutants have a reduced sucrose intake. (Li et al., 2016; Scheiner et al., 2014) To investigate whether starvation influences the reallocation of an external food cue, we starved flies and determined the preference to consume sucrose to protein-enriched food using the capillary feeder assay (CAFE) (Ja et al., 2007). After starvation, control flies and $T\beta h^{nM18}$ mutants chose between 5% sucrose and 5% yeast (Figure 3B). Prolonged starvation resulted in a decreased sucrose preference in controls and $T\beta h^{nM18}$ mutants, but $T\beta h^{nM18}$ mutants started with a lower sucrose preference. Only after 40 h of starvation did $T\beta h^{nM18}$ mutants show a similar preference to control sucrose consumption. To further investigate whether $T\beta h^{nM18}$ mutants have defects in the regulation of internal sucrose homeostasis, we deprived male flies of sucrose- or protein-enriched food by letting them feed on 5% sucrose, 5% yeast or standard food for 3 days and analyzed their food preference after deprivation (Figure 3C). As controls, we included mated females in our analysis since they have different nutritional requirements due to their mating status. (Ribeiro and Dickson, 2010; Vargas et al., 2010) Flies fed standard food for 3 days strongly preferred sucrose to protein-enriched food, with $T\beta h^{nM18}$ mutants showing a significant reduction. The yeast-deprived $T\beta h^{nM18}$ mutant showed a significantly reduced preference for sucrose, and sucrose-deprived mutants had sucrose preferences similar to those of controls. In mated females, no differences in food

preference between controls and the $T\beta h^{nM18}$ mutant were observed. In summary, the reduced preference to consume sucrose correlated in $T\beta h^{nM18}$ with increased glycogen levels. In addition, $T\beta h^{nM18}$ mutants can sense the reduction in specific internal energy supplies and change their food preferences accordingly.

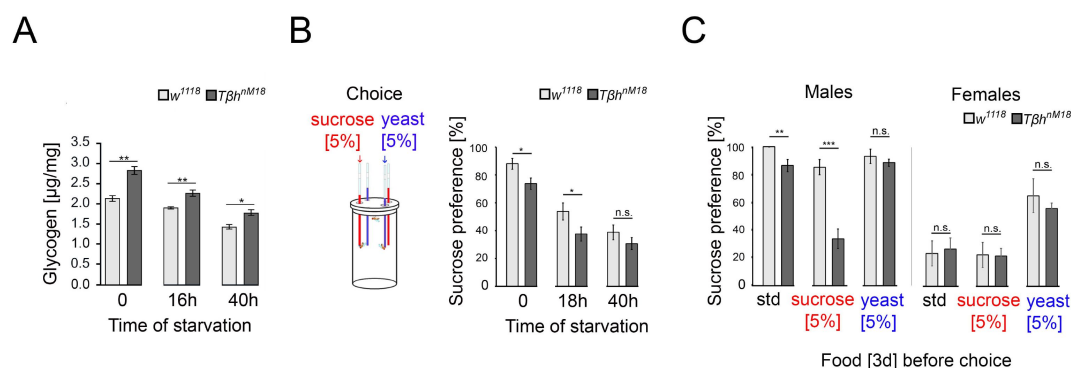


Figure 3.

Elevated glycogen levels correlate with reduced sucrose preference

A, Analysis of whole-body glycogen levels in w^{1118} and $T\beta h^{nM18}$ flies. Glycogen levels in $T\beta h^{nM18}$ flies are significantly higher than those in w^{1118} flies under similar starvation conditions. $N = 3$ groups of 5 male flies. **B**, Flies were starved 18 h or 40 h before food intake was measured for 24 h. Flies chose between 5% sucrose and 5% yeast. The preference was determined. All flies showed a significant preference for sucrose consumption. Starvation reduced the preference. $T\beta h^{nM18}$ showed a significantly reduced preference for sucrose after 18 h. $N = 20$ –26 groups of eight flies. **C**, Feeding flies for 3 days on standard, 5% sucrose or 5% yeast resulted in control flies preferring to consume sucrose. $T\beta h^{nM18}$ mutants fed normal food and sucrose showed a significant reduction in sucrose preference but not when fed 5% yeast for three days. $N = 20$ –28 groups of eight flies. To determine differences between two groups, Student's t test was used. $P^* < 0.05$; $P^{**} < 0.01$, $P^{***} < 0.001$.

Internal glycogen storage influences sucrose-related memories

Since memory performance increases upon reduction of the internal energy supplies due to starvation, we wanted to investigate whether the internal energy supply influences memory performance. In *Drosophila*, glycogen is mainly found in the fat bodies - the major energy storage organ - and the muscles, a major site of energy expenditure (Wigglesworth, 1949). Glycogen synthase and glycogen phosphorylase control the levels of glycogen in the body (Figure 4A). Knockdown of glycogen synthase using $GlyS^{HMS01279}$ -RNAi efficiently reduced glycogen levels in larvae, whereas knockdown of glycogen phosphorylase using $GlyP^{HMS00032}$ -RNAi efficiently increased glycogen levels (Yamada et al., 2018). We altered glycogen levels in the muscles using the $mef2$ -Gal4 driver (Ranganayakulu et al., 1998) and/or the fat bodies using the FB-Gal4 driver (Gronke et al., 2003) and analyzed the effect of changed glycogen levels on short-term memory (Figure 4). We used PAS staining to confirm the down- or upregulation of glycogen in larval muscles or fat bodies, respectively (Figure 4).

(Yamada et al., 2018) Additionally, we quantified the glycogen levels in the body of adult flies (Figure S2).

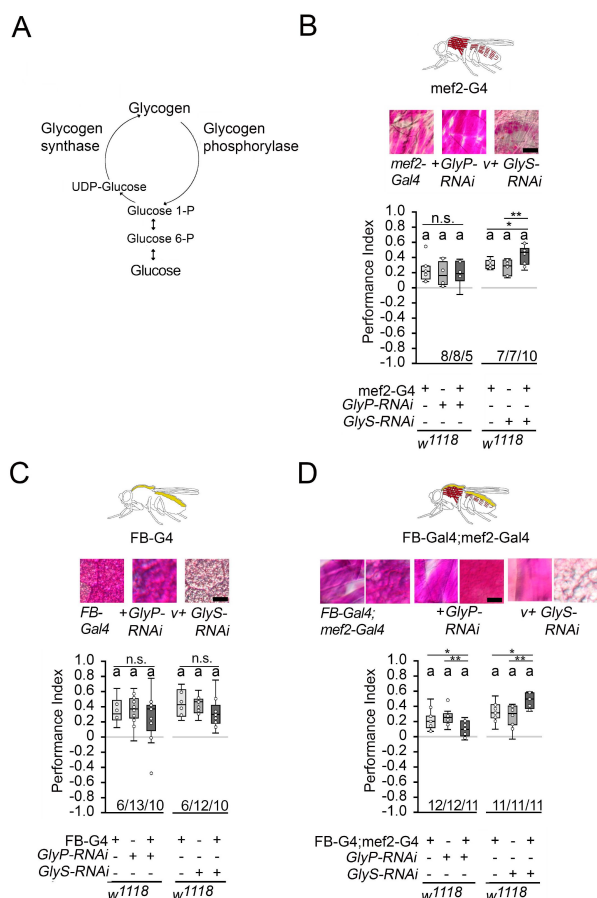


Figure 4.

Carbohydrate storage influences appetitive STM

A, Schema of glycogen synthesis. The expression of *GlyP-RNAi* reduced glycogen phosphorylase and increased glycogen levels, whereas *GlyS-RNAi* reduced glycogen synthase and decreased glycogen levels in target tissues. **B-D**, PAS was used to visualize glycogen levels in larval muscle or fat bodies. **B**, Increases in glycogen in the muscles have no effect on STM, whereas reduced muscle glycogen increases appetitive STM. **C**, Increased or decreased glycogen levels in the fat bodies did not interfere with STM. **D**, A combined increase in glycogen in muscles and fat bodies reduced STM, and a decrease in glycogen increased STM. Student's t tests were used to determine differences between two groups, and one-way ANOVA with post hoc Tukey's HSD was used to determine differences between three or more groups. The letter "a" marks a significant difference from random choice as determined by a one-sample sign test ($P^* < 0.05$; $P^{**} < 0.01$). Numbers below box blots indicate one pair of reciprocally trained independent fly groups.

Increasing glycogen levels in the muscles did not change short-term memory in 16 h starved flies, but the reduction in glycogen significantly improved memory strength (Figure 4B). Increasing or decreasing glycogen levels in the fat bodies had no effect on memory performance (Figure 4C). When the glycogen levels were significantly increased in the muscles and fat bodies, flies showed a reduced memory to odorants paired with sucrose. An increase in memory performance was observed when glycogen levels were significantly reduced in both tissues (Figure 4D). Recently, it has been shown that energy metabolism in mushroom bodies is important for the formation of LTM. (Placais et al., 2017) To analyze the function of the mushroom bodies, the expression of *GlyS^{HMS01279}-RNAi* under the control of the *Mef2-Gal4* driver was repressed in the mushroom bodies using the *mb247-Gal80* driver. (Krashes et al., 2007) The memory was still increased (Figure S3). Reciprocally, the reduction of *GlyS^{HMS01279}-RNAi* using the *mb247-Gal4* driver targeting the mushroom bodies (Zars et al., 2000) did not change short-term memory (Figure S3). Thus, low levels of glycogen in the muscles upon starvation positively influence appetitive short-term memory, while high levels of glycogen in the muscles and fat body reduce short-term memory.

Internal glycogen levels reduce sucrose-related memories in *Tβh* mutants

The elevated glycogen levels in *Tβh^{nM18}* mutants might be responsible for the reduced STM. To determine whether the reduction in glycogen levels in the muscles or fat bodies restores

STM, we expressed *GlyS^{HMS01279}-RNAi* under the control of the *mef2-Gal4* or *FB-Gal4* driver in *Tβh^{nm18}* mutants and analyzed STM (Figure 5A). Neither the reduction in the muscles nor the reduction in fat bodies of *Tβh^{nm18}* mutants improved STM. Only when glycogen was reduced in both tissues did the *Tβh^{nm18}* mutants show improved STM compared to controls. Thus, *Tβh^{nm18}* mutant flies can form appetitive STM similar to controls when energy storage is sufficiently reduced. Next, we analyzed whether male *Tβh^{nm18}* mutants can form STM to other nutrients than carbohydrates by using a protein-enriched diet in the form of 5% yeast as a positive reinforcer (Figure 5B). To evaluate whether there is a difference in the evaluation of protein as a food source between male flies and *Tβh^{nm18}* mutants, we determined yeast intake in non-starved and starved flies (Figure S4). Non-starved *Tβh^{nm18}* mutant males have a significantly higher protein intake than controls. However, after 16 h of starvation, the level of protein intake was comparable to that of the controls. Using 5% yeast as a food reward, male *Tβh^{nm18}* mutants showed comparable levels of STM to controls (Figure 5B).

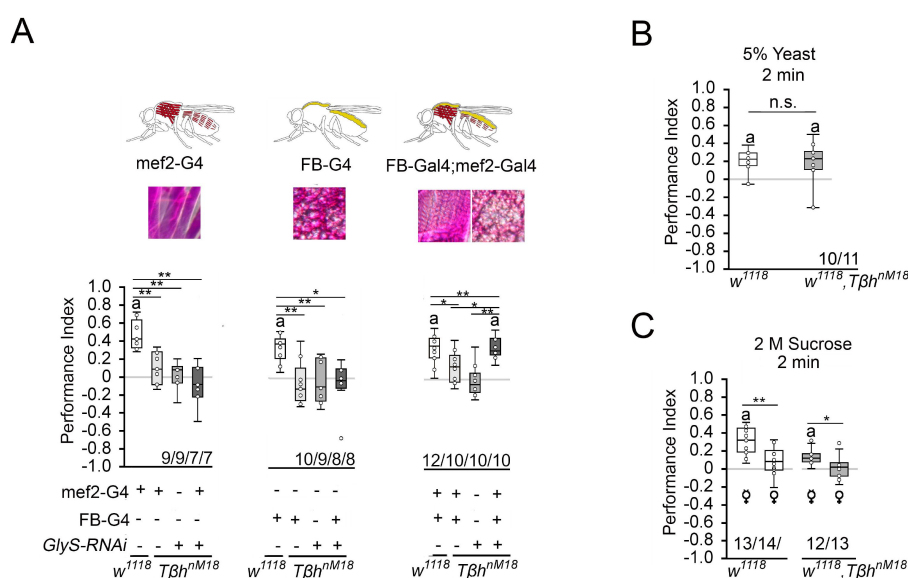


Figure 5.

Reducing glycogen in *Tβh^{nm18}* improves appetitive STM

A, Decreasing glycogen concentration using *UAS-GlyS^{RNAi}* in the muscles or fat bodies in *Tβh^{nm18}* mutants did not improve STM, but decreasing glycogen in both tissues improved STM to control levels. **B**, *w¹¹¹⁸* and *Tβh^{nm18}* flies formed similar levels of appetitive STM when 5% yeast was used as a reinforcer. **C**, Virgin females of *w¹¹¹⁸* and *Tβh^{nm18}* displayed STM, whereas mated females of both genotypes did not.

Differences between two groups were determined using Student's t tests, and differences among more than two groups were determined with one-way ANOVA with Tukey's HSD post hoc test. Differences from random choice were determined using a one-sample sign test and marked with the letter "a". $P^* < 0.05$; $P^{**} < 0.01$. Numbers below box blots indicate one pair of reciprocally trained independent fly groups.

To further analyze the influence of the internal energy status on memory performance, we took advantage of the observation that female flies have different nutritional requirements depending on their mating status. (Ribeiro and Dickson, 2010; Vargas et al., 2010) Virgin females showed a higher consumption preference for sucrose than mated females when given the choice between yeast and sucrose (Figure 3C), supporting a difference in internal energy demands and suggesting a difference in valence for different diets. We tested virgin and mated female flies of controls and *Tβh^{nm18}* mutants for STM (Figure 5B). Virgin females remembered sucrose significantly better than mated females. This was also true for *Tβh^{nm18}* mutant females. Thus, the internal energy status influences how the reward is evaluated in learning and memory processing.

Insulin-like signaling in octopaminergic neurons regulates STM

How is the internal energy status integrated into the reward system? Insulin-like signaling regulates glycogen levels in invertebrates and vertebrates.(Semaniuk et al., 2021) Loss of the insulin receptor results in more circulating sugar but not increased glycogen levels. (Shingleton et al., 2005) Thus, the insulin receptor might be a good candidate that links the internal energy level to reinforcing neurons. First, we analyzed whether the insulin receptor is expressed in octopaminergic reward neurons in the brain (Figure 6). To detect the expression of an activated insulin receptor, we used an insulin antibody that recognizes the phosphorylated form of the insulin receptor (InR). This region is highly conserved between humans and flies. First, we tested whether the antibody indeed recognizes the activated insulin receptor. Therefore, we overexpressed the activated insulin receptor using the UAS-InR.A1325D transgene under the control of the *dTdc2*-Gal4 driver (Figure S5). The InR.A1325D protein variant mimics the human V938D protein variant that is constitutively active,(Longo et al., 1992) and the *dTdc2*-Gal4 driver targets octopaminergic reward neurons. (Busch et al., 2009) The expression of activated InR resulted in increased immune reactivity (Figure S5). Thus, the InR antibody detects the activated InR. In the brain, activated InR is broadly expressed in a punctate manner and more specifically in the soma of *dTdc2*-Gal4-targeted neurons (Figure 6A). To uncouple energy sensing via the insulin receptor in reward neurons, we expressed UAS-InR.K1409A under the control of the *dTdc2*-Gal4 driver. The transgene encodes a dominant negative variant of the InR (InR^{DN}) and interferes with InR function.(Wu et al., 2005) In 16 h starved animals, the expression of InR^{DN} under the control of the *dTdc2*-Gal4 driver did not change STM, but uncoupling of InR-dependent energy sensing in reward neurons in *Tβh^{nM18}* restored STM to normal levels (Figure 6B and C).

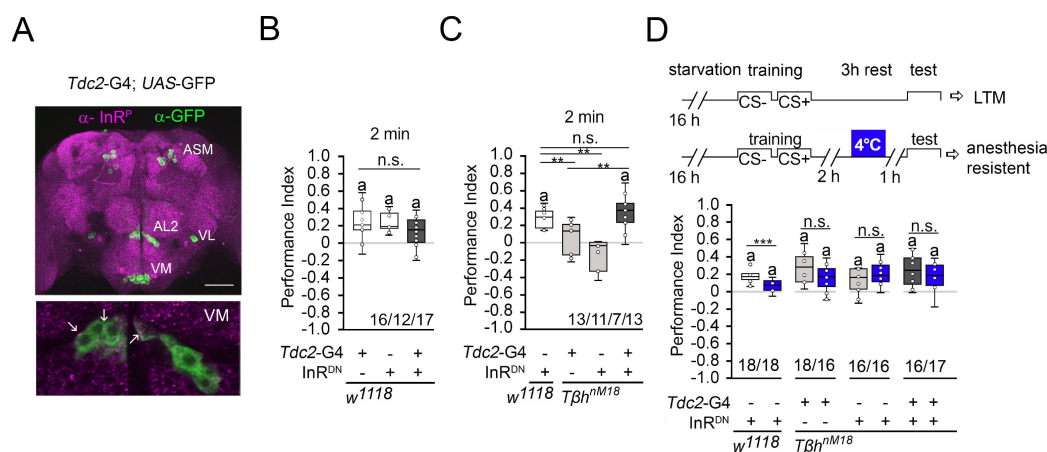


Figure 6.

Insulin signaling in reward neurons regulates STM performance

A, The activated form of the InR is expressed in punctate manner throughout the brain (in magenta) and is also detected in octopaminergic reward neurons visualized by using the UAS-mcd8::GFP transgene under the control of the *Tdc2*-Gal4 driver (in green). **B**, Blocking InR signaling in *Tdc2*-Gal4-targeted octopaminergic neurons does not change appetitive STM in 16 h starved flies. **C**, Blocking InR signaling in *Tdc2*-Gal4-targeted octopaminergic neurons in *Tβh^{nM18}* mutants restored STM to control levels. **D**, A cold shock did not disrupt emerging memory in *Tβh^{nM18}* mutants with blocked InR under the control of the *Tdc2*-Gal4 driver.

Student's *t* test was used to determine differences between two groups, and one-way ANOVA with Tukey's post hoc HSD test was used to determine differences between three or more groups. The letter "a" marks a significant difference from random choice as determined by a one-sample sign test ($P < 0.05$). n.s. is not significant; $P^* < 0.05$; $P^{**} < 0.01$.

To investigate whether the improved STM also affects the emerging LTM of the $T\beta h^{nM18}$ mutants, we performed cold-shock experiments in $T\beta h^{nM18}$ mutants in which InR^{DN} was expressed in octopaminergic reward neurons (Figure 6D). The 3 h memory in controls is cold-shock sensitive, but the memory in $T\beta h^{nM18}$ mutants is cold-shock insensitive, supporting the idea that the mutants formed anesthesia-resistant memory. Given that octopamine is a negative regulator of memory and that it is still missing in $T\beta h^{nM18}$ mutants with blocked insulin signaling on octopaminergic reward neurons, it is not surprising that ARM is still observed in the mutants. Thus, two functionally distinct memory traces were formed after training, an insulin receptor-sensitive appetitive short-term memory and a long-term memory that can be blocked by octopamine.

Increased starvation results in overconsumption in $T\beta h^{nM18}$ mutants

To analyze whether the increased glycogen levels and the reduced sucrose reward correlate with food consumption, the energy demands in flies under different starvation conditions were analyzed by measuring food intake (Figure 7). When control flies were starved for 16 h or 40 h, they consumed similar amounts of 5% sucrose. In contrast, 16 h-starved $T\beta h^{nM18}$ mutants consumed significantly less sucrose, but after 40 h of starvation, they consumed approximately 34% more sucrose (Figure 7B). After 40 h of starvation, the glycogen levels in $T\beta h^{nM18}$ mutants were still higher than those in controls (Figure 3A). Thus, they overconsumed sucrose. The overconsumption was independent of the diet, as they showed similar overconsumption when fed with a solution containing 5% yeast and 5% sucrose (Figure 7B). To investigate whether the integration of the internal energy status is also integrated into feed behavior by octopaminergic reward neurons, we blocked insulin signaling in reward neurons in 16 h-starved $T\beta h^{nM18}$ cells and analyzed sucrose consumption (Figure 7C). The reduced sucrose consumption of $T\beta h^{nM18}$ cells was significantly improved to control levels when insulin signaling was blocked in reward neurons. Thus, the regulation of food consumption also requires the integration of internal energy levels into the reward system via octopaminergic neurons.

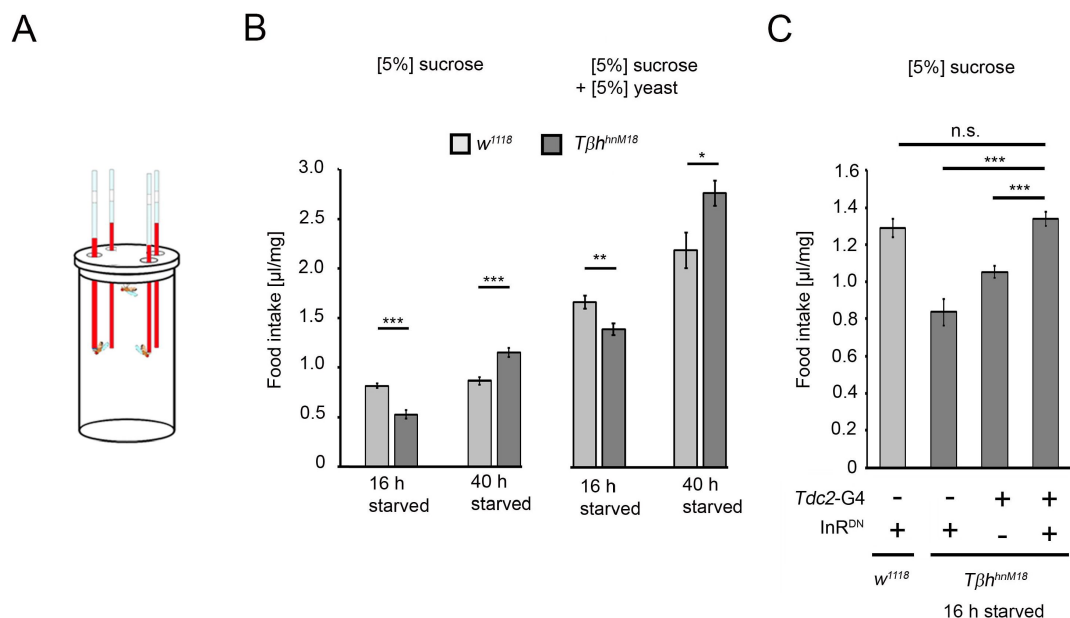


Figure 7.

Prolonged starvation results in rebound sucrose intake in hyperglycemic *Tβh^{NM18}* mutants

A, Capillary feeder assay used to determine food intake. **B**, Flies were starved 16 h or 40 h before 24 h food intake was measured. After 16 h of starvation, *Tβh^{NM18}* mutants significantly consumed less 5% sucrose or 5% sucrose with 5% yeast, and after 40 h of starvation, *Tβh^{NM18}* mutants significantly consumed more sucrose. N = 20–26 groups of eight flies. **C**, Blocking InR signaling in *Tdc2-Gal4*-targeted octopaminergic neurons in *Tβh^{NM18}* mutants significantly increased 5% sucrose consumption. N = 20–28 groups of eight flies. To determine differences between two groups, Student's t test was used, and to determine differences between three or more groups, one-way ANOVA with post hoc Tukey's HSD was used. $P^* < 0.05$; $P^{**} < 0.01$, $P^{***} < 0.001$.

Discussion

Depending on the internal energy level, flies form first protein-dependent aversive olfactory LTM and later ARM, a protein synthesis-independent form of memory (Wu et al., 2005). In times of energy shortage, for the organism, costly protein biosynthesis-dependent LTM is exchanged against “less costly” protein biosynthesis-independent ARM (Mery and Kawecki, 2005). Similarly, we observed that after shorter periods of starvation, appetitive LTM appeared, whereas after longer periods of starvation, ARM emerged (Figure 1E). The *Tβh^{NM18}* mutants form only anesthesia-sensitive aversive memory (Wu et al., 2013). In contrast, depending on the duration of starvation, the emerging appetitive memory in *Tβh^{NM18}* mutants is cold-shock sensitive or cold-shock insensitive. Thus, *Tβh^{NM18}* mutants are not defective in appetitive ARM, and the internal energy status defines how quickly ARM is formed.

Octopamine gates memory formation

What is the function of octopamine in memory formation? Octopamine is a negative regulator of emerging LTM, as feeding octopamine receptor antagonists to control flies results in memory, and emerging memory in *Tβh^{nM18}* mutants is suppressed by octopamine after conditioning. In addition, octopamine acts upstream of dopaminergic neurons in the formation of LTM, since blocking neurotransmitter release of dopaminergic LTM-mediating neurons directly after conditioning blocks memory formation in *Tβh^{nM18}* mutants. The results are consistent with the model that dopamine release is not required during the pairing of the conditioned stimulus and the reward rather than as gain control after the training. (Adel and Griffith, 2021) The gain can be controlled by the inhibitory function of octopamine.

Similar to the function of octopamine upstream of dopaminergic neurons in LTM, octopamine acts upstream of dopaminergic neurons during the acquisition of appetitive STM. (Burke et al., 2012; Liu et al., 2012) However, at first glance, the function of octopamine seems to differ in STM and LTM. Loss of octopamine in *Tβh^{nM18}* mutants results in loss of STM, supporting that octopamine is required as a positive regulator for appetitive STM. (Schwaerzel et al., 2003) The loss of STM in *Tβh^{nM18}* mutants has been attributed to the loss of labile STM that forms in response to the sweetness of sucrose but not due to defects in stable LTM that forms in response to the caloric value of sugar. (Burke et al., 2012; Fujita and Tanimura, 2011) Sweetness is sensed by gustatory receptor neurons that change their sensitivity upon starvation (Inagaki et al., 2012; Marella et al., 2012) and thereby influence responsiveness to the external reinforcer. Responsiveness might be regulated by the octopaminergic system, since octopaminergic neurons regulate the sucrose sensitivity of gustatory neurons in *Drosophila*. (Youn et al., 2018) However, the reduced sensitivity of gustatory sucrose receptors cannot explain why octopamine-deficient mutants do not form STM, since *Tβh^{nM18}* mutants form aversive memory to lower sucrose concentrations.

The internal energy status influences how the external source of sucrose is evaluated rather than octopamine being directly involved in the acquisition of appetitive STM. Consistently, *Tβh^{nM18}* is able to show similar levels of STM as controls after longer periods of starvation. This is further strengthened by the results showing that octopaminergic reward neurons that are uncoupled from energy sensing by inactivation of insulin receptor signaling and reduced glycogen levels in *Tβh^{nM18}* flies also improve appetitive STM. In addition, *Tβh^{nM18}* mutant females with different energy requirements form appetitive STM, and a short pulse of octopamine before conditioning can even block STM. Thus, octopamine is not required in the acquisition of appetitive STM but rather suppresses LTM when enough energy is available. In addition, the function of OA as a suppressor of different forms of memory allows for differential selection of what kind of memory is formed depending on internal and external information. This gating function in memory formation by octopamine can also be observed in other behaviors. For example, octopamine regulates the decision to approach or avert a food source. (Classen and Scholz, 2018) Thus, the octopaminergic system integrates internal energy demands and the evaluation of external information.

The internal energy level of the animal influences memory formation

On the cellular level, insulin receptor signaling regulates energy metabolism. (Chatterjee and Perrimon, 2021) The broad expression pattern of the insulin receptor in the *Drosophila* brain (Figure 4) indicates that every cell in the brain needs to regulate its own energy homeostasis. However, the overall energy resources of the fly and the evaluation of the external food supply also need to be integrated into memory formation. The external food supply might be

evaluated by the reward system. In mice, insulin receptor signaling in dopaminergic neurons mediates food reward.(Konner et al., 2011) Consistently, insulin receptor signaling in octopaminergic reward neurons regulates the rewarding properties of sucrose in appetitive STM memory (Figure 4C) and food intake (Figure 5C).

The systemic metabolic rate is regulated by fat bodies. For example, after food intake, the fat body secretes Unpaired 2 (Upd2), which in turn regulates the secretion of Dilps via GABergic neurons.(Rajan and Perrimon, 2012) Reduction of glycogen in the fat bodies is not sufficient to change appetitive STM, but reduction in the muscles or both tissues is. The results support a feedback mechanism between the evaluation of glycogen levels in the muscles and the brain. In addition, the muscles and the fat bodies communicate about the energy content of both structures, and this information is signaled back to the brain. Such long-range signals from muscles exist. Skeletal muscles are secretory organs.(Pedersen and Febbraio, 2012) For example, the muscle-secreted Amyrel amylase reduces the age-related accumulation of polyubiquitinated proteins in the brain.(Rai et al., 2021) Independent of how the glycogen levels of both tissues are communicated to the brain, they influence how the reinforcer is evaluated in learning and memory processing. Feedback between the energy levels in both structures is still intact in octopamine-deficient mutants, since T β h^{nM18} mutants still form STM when the energy storage is sufficiently reduced.

Are food-related memories and internal glycogen levels predictive of food intake?

Feeding behaviors in hungry animals are regulated by different neural systems, including networks receiving and processing sensory information and networks assigning reward properties to food.(Berridge, 2009) Blocking insulin receptor function in dopaminergic neurons causes increased food intake and weight gain in mice.(Konner et al., 2011) Similarly, blocking insulin receptor function in octopaminergic reward neurons in flies with increased glycogen levels and reduced food intake increases food intake. Thus, the octopaminergic system assigns reward properties to food. However, blocking insulin receptor signaling on reward neurons does not result in extensive overconsumption, supporting that the regulation of the food amount is still intact and is not due to insulin receptor resistance in reward neurons. Increased levels of internal energy, e.g., glycogen, reduced the reward properties of food, resulting in a decreased positive association of food-related memories and decreased food consumption. This should prevent increased food intake in the presence of excess food. Insulin resistance might contribute to weight gain, since in addition to high levels of glycogen, the flies show normal food intake. However, weight gain might be differentially regulated by overeating. The overconsumption is only visible after a prolonged phase of starvation that is also correlated with emerging stable appetitive long-term memory. This suggests that the stability of appetitive memory might result in the re-evaluation of food and might trigger the overconsumption of food.

In summary, the octopaminergic neurotransmitter system integrates several aspects that influence the regulation of food memories. Octopamine modules the sensory perception of the conditioned stimulus and the sensory perception of the reward. Here, we show that the evaluation of food reward in the context of energy storage is integrated by the octopaminergic system and influences the stability of food-related memories. The function of octopamine as a negative regulator of different forms of memory allows for selective regulation of associated behaviors such as food intake, and the loss of this regulation might also promote dysregulation of food intake. The close relationship between the octopaminergic neurotransmitter and the noradrenergic neurotransmitter system suggests that integrator function might be conserved.

Material and methods

Drosophila melanogaster

Flies were raised on an ethanol-free standard cornmeal-molasses food at 25°C and 60% relative humidity on a 12 h/12 h day-night cycle. The following lines were used: *w¹¹¹⁸*; *w¹¹¹⁸, Tβh^{nM18}*; (Monastirioti et al., 1996) *UAS-GlyS^{HMS01279}-RNAi* (BDSC #34930); *UAS-GlyP^{HMS00032}-RNAi* (BDSC #33634); *w¹¹¹⁸*; *UAS-shi^{ts}*; (Kitamoto, 2001) *w¹¹¹⁸*; P{*UAS-InR*. K1409A}, (BDSC#8253); *w¹¹¹⁸*; *FB-Gal4* (a generous gift from the Partridge Lab); *mef2-Gal4*; (Ranganayakulu et al., 1998) *R15A04-Gal4* (BDSC #48671); *Tdc2-Gal4*. (Cole et al., 2005) All lines were backcrossed to *w¹¹¹⁸* (Scholz Lab) for at least five generations to isogenize the genetic background. For behavioral experiments, three- to five-day-old male flies were used, if not otherwise indicated. Male flies of the F1 generation carrying one copy of the transgene were used as controls for behavioral experiments. Animal studies using *Drosophila melanogaster* were conducted in agreement with the regulations of the DFG and the Land North Rhine-Westphalia. Other ethics approval and informed consent statement are not applicable for research using *Drosophila melanogaster*.

Olfactory learning and memory

Associative olfactory learning and memory was trained and tested with a modified version of the Tully and Quinn olfactory conditioning apparatus. (Schwaerzel et al., 2003; Tully and Quinn, 1985) Approximately 70 one- to two-day-old male flies were collected with CO₂ anesthesia and were kept for 2 days at 25°C to recover from CO₂ sedation. Briefly, three- to five-day-old male flies were starved for 16 h or 40 h in vials with water-soaked filter paper at the bottom. Flies were transferred to training tubes and exposed to the first odorant for 2 min, either 3-octanol (3-OCT diluted 1:80 in paraffin oil) or 4-methylcyclohexanol (MCH diluted 1:100 in paraffin oil). After that, they were transferred to a second tube and exposed to the second odorant in the presence of filter paper soaked with either 0.15 M sucrose, 2 M sucrose or 5% yeast. To analyze the abilities of the flies to learn and remember the odorant paired with the reward, 2 min, 3 h or 6 h after training, flies were given the choice between odorant 1 (CS+) and odorant 2 (CS-) for 2 min. The performance index (PI) was calculated as $PI = (\#(CS+) + \#(CS-)) / (\text{total } \# \text{ flies})$, where CS+ indicates the odorant associated with the appetitive reinforcer and CS- indicates the odorant that was not associated with the reinforcer. To exclude nonassociative effects, each “n” consists of a reciprocally trained, independent group of naïve flies. For pharmacological experiments, flies were fed water, 3 mM OA or 2 M sucrose before training or with 3 mM OA or 3 mM Epinastine between training and testing. The assay was performed at RT with 60% relative humidity. For cold-shock experiments, flies in vials were placed in ice-cold water for 2 min. Odorant acuity and odorant balance were tested with naïve flies placed into the Tully Quinn setup. They chose for 2 min between odorant- or paraffin oil-containing sides or between both odorants. For sucrose and yeast preference, flies chose between sucrose- or yeast-soaked filter paper and water-soaked filter paper.

Food intake

To analyze the consumption of nutrients in flies, the capillary feeder (CaFe) assay was performed. (Diegelmann et al., 2017) Briefly, eight three- to five-day-old male or female flies that were either nonstarved or starved for 18 h or 40 h had access to 4 capillaries filled with either 5% sucrose or 5% yeast or a mixture of both for 24 h at 25°C and 60% humidity. The solution was colored with red food dye. During starvation, flies were kept in vials with wet

filter paper at 25°C and 60% humidity. The amount of consumed solution was determined using an electronic caliper. To account for evaporation, the average evaporation of three CAFE setups without flies was measured and used to normalize the average food intake per fly. To normalize food intake to body weight, at least 5 times 100 male flies per genotype and condition were weighed, and the average body weight of a single fly was determined. The total consumption per μg fly was calculated by dividing the total consumption per fly by the mean weight of a fly. N indicates the number of tested groups.

Glycogen content

Whole body glycogen levels were determined with the Glucose (HK) Assay Kit (Sigma Aldrich, #GAHK20-1KT) according to the protocol of (Tennesen et al., 2014). A group of five male flies that were either sated or starved for 16 h or 40 h were homogenized in 100 μl ice-cold 1x PBS. To reduce enzymatic degradation, proteins and enzymes were heat inactivated at 70°C for 10 min. The supernatant was removed and diluted 1:3 with 1x PBS. Twenty microliters of the samples were either added to 20 μl of 1x PBS or PBS/Amyloglucosidase mix and incubated at 37°C for 60 min. Then, 100 μl of HK-reagent was added to the sample or glucose standard and incubated for 15 min at RT. Absorbance was measured at 340 nm. The glycogen content was calculated by subtracting the total glucose concentration from 1x PBS-treated samples from the total glucose of amyloglucosidase-treated samples.

Periodic acid staining

To visualize glycogen levels in the fat bodies and muscle tissue of larvae and adult *Drosophila*, PAS staining was performed after Yamada et al. (Yamada et al., 2018) with a slight modification. The samples were fixed in 3.5% formaldehyde for 20 min and washed two times for 5 min with 1% BSA/PBS. Periodic acid solution was added for 5 min, followed by two washes of 5 min with 1% BSA/PBS. Schiff's Reagent was added for 5 min, followed by two washes of 5 min with 1% BSA/PBS. Tissue was stored in 50% glycerol.

Immunohistochemistry

For immunohistochemistry, antibodies raised in rabbits against the activated insulin-like receptor (cell signaling technology #3021) were used at a dilution of 1:50 in 5% normal goat serum in PBS with 0.1% Triton and incubated for two days. The brains were washed with PBS with 0.3% Triton.

Quantification and statistical analysis

Food intake was displayed as the mean $\pm$ s.e.m. For learning and memory experiments, the data were displayed as boxplot $\pm$ minimum ($Q1 - 1.5 \times \text{IQR}$) and maximum ($Q1 + 1.5 \times \text{IQR}$). The nonparametric one-sample sign test and the parametric one-sample t test were used to analyze whether behavior was based on random choice. Differences between two groups were determined with Student's t test, and more groups were compared with one-way ANOVA with Tukey's post hoc HSD test. Statistical analysis was performed with Statistica 9.1 (StatSoft, Tulsa, OK, USA). Boxplots were generated with Microsoft Excel 2016 and GIMP 2.10.12.

Data availability

All data related to figures are included in the supplement Table S2.

Acknowledgements

We thank the Linda Partridge lab and the Bloomington *Drosophila* Stock Center for providing *Drosophila* fly lines. H.S. was supported by SFB1340, DFG-Scho656-10-1 and the Hetzler award.

Author contributions

H.S. initiated the project. M.B., H.S. and K.D. designed the experiments. M.B., K.A., K. D. and T.E.K. performed the experiments and, together with H.S., analyzed the data. M.B. and K.A. performed learning and memory tests. K.D. characterized insulin receptor expression. M.B. and T.E.K. performed food consumption assays. M.B. wrote a draft of the manuscript, and H.S. wrote the manuscript.

Declaration of interests

The authors declare no competing interests.

Supplemental information titles and legends

Supplemental information includes 5 figures and two tables and can be found with this article online.

Figure S1. Starvation influences memory performance using 0.15 M sucrose as a reinforcer

A, Appetitive 2 min STM using 0.15 M sucrose as a reinforcer of w^{1118} and $T\beta h^{nM18}$ mutants starved for 16 h (white bars) or 40 h (gray bars) before the training. After 16 h of starvation, w^{1118} mutants formed STM, and prolonged starvation increased STM. After 16 h of starvation, $T\beta h^{nM18}$ mutants formed aversive STM to 0.15 M sucrose and prolonged starvation resulted in appetitive STM. Numbers below box blots indicate one pair of reciprocally trained independent groups of male flies. Student's t tests were used to determine differences between two groups, and one-way ANOVA with Tukey's post hoc HSD test was used to determine differences between three groups. The letter "a" marks a significant difference from random choice as determined by a one-sample sign test ($P^* < 0.05$; $P^{**} < 0.01$).

Figure S2. Glycogen level in adult flies with reduced GlyP and GlyS

The relative glycogen levels in flies with altered GlyP and GlyS in muscles, fat bodies or both. The expression of *GlyP-RNAi* and *GlyS-RNAi* under the control of *mef2-Gal4* did not significantly change the relative glycogen level but did under the control of the FB-Gal4

driver. The expression of *GlyP-RNAi* in fat and muscle tissue resulted in significantly increased glycogen levels, and the expression of *GlyS-RNAi* significantly decreased glycogen levels. The glycogen was measured in 3 groups of 5 male flies and normalized to the protein levels and to the glycogen levels of flies of the Gal4 driver. ANOVA with post hoc Tukey's HSD was used to determine differences between three groups. ($P^* < 0.05$; $P^{**} < 0.01$).

Figure S3. The glycogen level in the mushroom bodies does not influence appetitive STM

A, The expression of *GlyS-RNAi* under control of the *mef2-Gal4*; *mb247-Gal4* drivers increased STM. **B**, The expression of *GlyS-RNAi* under control of the *mb247-Gal4* driver did not change STM. One-way ANOVA with Tukey's HSD post hoc test was used to determine differences between the groups. The letter "a" marks a significant difference from random choice as determined by a one-sample sign test ($P^* < 0.05$; $P^{**} < 0.01$; n.s. = non-significant). Numbers below box blots indicate one pair of reciprocally trained independent fly groups. **C**, Sensory acuity tests for the genotypes used.

Figure S4. Starvation influences yeast consumption of *Tβh^{nm18}* flies

Non-starved *Tβh^{nm18}* male flies consumed significantly more yeast for 24 h. After 16 h of starvation, *Tβh^{nm18}* mutants consumed similar amounts of yeast within 24 h. N = 26 groups of eight flies. To determine differences between both groups, Student's t test was used. $***P < 0.001$; n.s. = nonsignificant difference.

Figure S5. The antibody against phosphorylated InR recognizes activated InR

In the antennal lobes (AL) of the male adult brain, the immunoreactivity recognized by the anti-InR^P antibody is shown in magenta, and GFP expression is shown in green. **A**, **B**, Gal4 expression of the *Tdc2-Gal4* driver targeting octopaminergic neurons is visualized using the UAS-mCD8::GFP transgene. **B**, The expression of the constitutively active insulin receptor in octopaminergic neurons results in increased immunoreactivity detected by the anti-InR^P antibody.

Table S1. Sensory acuity

Table S2. Data related to figures

References

- Adel M. , Griffith L.C (2021) **The role of dopamine in associative learning in *Drosophila*: an updated unified model** *Neurosci Bull* **37**:831–852
- Berridge K.C (2009) **'Liking' and 'wanting' food rewards: brain substrates and roles in eating disorders** *Physiol Behav* **97**:537–550
- Burke C.J. , Huetteroth W. , Oswald D. , Perisse E. , Krashes M.J. , Das G. , Gohl D. , Silies M. , Certel S. , Waddell S (2012) **Layered reward signalling through octopamine and dopamine in *Drosophila*** *Nature* **492**:433–437
- Busch S. , Selcho M. , Ito K. , Tanimoto H (2009) **A map of octopaminergic neurons in the *Drosophila* brain** *J Comp Neurol* **513**:643–667

- Chambers D.B. , Androschuk A. , Rosenfelt C. , Langer S. , Harding M. , Bolduc F.V (2015) **Insulin signaling is acutely required for long-term memory in *Drosophila*** *Front Neural Circuits* **9**
- Chatterjee N. , Perrimon N (2021) **What fuels the fly: energy metabolism in *Drosophila* and its application to the study of obesity and diabetes** *Sci Adv* **7**
- Classen G. , Scholz H (2018) **Octopamine shifts the behavioral response from indecision to approach or aversion in *Drosophila melanogaster*** *Front Behav Neurosci* **12**
- Cole S.H. , Carney G.E. , McClung C.A. , Willard S.S. , Taylor B.J. , Hirsh J (2005) **Two functional but noncomplementing *Drosophila* tyrosine decarboxylase genes: distinct roles for neural tyramine and octopamine in female fertility** *J Biol Chem* **280**:14948–14955
- Colomb J. , Kaiser L. , Chabaud M.A. , Preat T (2009) **Parametric and genetic analysis of *Drosophila* appetitive long-term memory and sugar motivation** *Genes Brain Behav* **8**:407–415
- Damrau C. , Toshima N. , Tanimura T. , Brembs B. , Colomb J (2017) **Octopamine and tyramine contribute separately to the counter-regulatory response to sugar deficit in *Drosophila*** *Front Syst Neurosci* **11**
- Das G. , Klappenbach M. , Vrontou E. , Perisse E. , Clark C.M. , Burke C.J. , Waddell S (2014) ***Drosophila* learn opposing components of a compound food stimulus** *Curr Biol* **24**:1723–1730
- Diegelmann S. , Jansen A. , Jois S. , Kastenholz K. , Escarcena L.V. , Strudthoff N. , Scholz H (2017) **The CAPillary FEeder assay measures food intake in *Drosophila melanogaster*** *J Vis Exp*
- Ellingwood S.S. , Cheng A (2018) **Biochemical and clinical aspects of glycogen storage diseases** *J Endocrinol* **238**:R131–R141
- Fujita M. , Tanimura T (2011) ***Drosophila* evaluates and learns the nutritional value of sugars** *Curr Biol* **21**:751–755
- Galikova M. , Klepsatel P (2023) **Endocrine control of glycogen and triacylglycerol breakdown in the fly model** *Semin Cell Dev Biol* **138**:104–116
- Grillo C.A. , Piroli G.G. , Lawrence R.C. , Wrighten S.A. , Green A.J. , Wilson S.P. , Sakai R.R. , Kelly S.J. , Wilson M.A. , Mott D.D. , Reagan L.P (2015) **Hippocampal insulin resistance impairs spatial learning and synaptic plasticity** *Diabetes* **64**:3927–3936
- Gronke S. , Beller M. , Fellert S. , Ramakrishnan H. , Jackle H. , Kuhnlein R.P (2003) **Control of fat storage by a *Drosophila* PAT domain protein** *Curr Biol* **13**:603–606

Huetteroth W. , Perisse E. , Lin S. , Klappenbach M. , Burke C. , Waddell S (2015) **Sweet taste and nutrient value subdivide rewarding dopaminergic neurons in *Drosophila*** *Curr Biol* **25**:751–758

Inagaki H.K. , Ben-Tabou de-Leon S. , Wong A.M. , Jagadish S. , Ishimoto H. , Barnea G. , Kitamoto T. , Axel R. , Anderson D.J (2012) **Visualizing neuromodulation in vivo: TANGO-mapping of dopamine signaling reveals appetite control of sugar sensing** *Cell* **148**:583–595

Inoue Y.H. , Katsube H. , Hinami Y (2018) ***Drosophila* models to investigate insulin action and mechanisms underlying human diabetes mellitus** *Adv Exp Med Biol* **1076**:235–256

Ja W.W. , Carvalho G.B. , Mak E.M. , De la Rosa N.N. , Fang A.Y. , Liong J.C. , Brummel T. , Benzer S. (2007) **Prandiology of *Drosophila* and the CAFE assay** *Proc Natl Acad Sci U S A* **104**:8253–8256

Kitamoto T (2001) **Conditional modification of behavior in *Drosophila* by targeted expression of a temperature-sensitive shibire allele in defined neurons** *J Neurobiol* **47**:81–92

Konner A.C. , Hess S. , Tovar S. , Mesaros A. , Sanchez-Lasheras C. , Evers N. , Verhagen L.A. , Bronneke H.S. , Kleinridders A. , Hampel B. , et al. (2011) **Role for insulin signaling in catecholaminergic neurons in control of energy homeostasis** *Cell Metab* **13**:720–728

Krashes M.J. , Keene A.C. , Leung B. , Armstrong J.D. , Waddell S (2007) **Sequential use of mushroom body neuron subsets during *Drosophila* odor memory processing** *Neuron* **53**:103–115

Krashes M.J. , Waddell S (2008) **Rapid consolidation to a radish and protein synthesis-dependent long-term memory after single-session appetitive olfactory conditioning in *Drosophila*** *J Neurosci* **28**:3103–3113

Li Y. , Hoffmann J. , Li Y. , Stephano F. , Bruchhaus I. , Fink C. , Roeder T (2016) **Octopamine controls starvation resistance, life span and metabolic traits in *Drosophila*** *Sci Rep* **6**

Liu C. , Placais P.Y. , Yamagata N. , Pfeiffer B.D. , Aso Y. , Friedrich A.B. , Siwanowicz I. , Rubin G.M. , Preat T. , Tanimoto H (2012) **A subset of dopamine neurons signals reward for odour memory in *Drosophila*** *Nature* **488**:512–516

Longo N. , Shuster R.C. , Griffin L.D. , Langley S.D. , Elsas L.J (1992) **Activation of insulin receptor signaling by a single amino acid substitution in the transmembrane domain** *J Biol Chem* **267**:12416–12419

Marella S. , Mann K. , Scott K (2012) **Dopaminergic modulation of sucrose acceptance behavior in *Drosophila*** *Neuron* **73**:941–950

Mery F. , Kaweckí T.J (2005) **A cost of long-term memory in *Drosophila*** *Science* **308**

- Monastirioti M. , Linn Jr C.E. , White K (1996) **Characterization of *Drosophila* tyramine beta-hydroxylase gene and isolation of mutant flies lacking octopamine** *J Neurosci* **16**:3900–3911
- Naganos S. , Horiuchi J. , Saitoe M (2012) **Mutations in the *Drosophila* insulin receptor substrate, CHICO, impair olfactory associative learning** *Neurosci Res* **73**:49–55
- Nakai J. , Chikamoto N. , Fujimoto K. , Totani Y. , Hatakeyama D. , Dyakonova V.E. , Ito E (2022) **Insulin and memory in invertebrates** *Front Behav Neurosci* **16**
- Pedersen B.K. , Febbraio M.A (2012) **Muscles, exercise and obesity: skeletal muscle as a secretory organ** *Nat Rev Endocrinol* **8**:457–465
- Placais P.Y. , De Treder E. , Scheunemann L. , Trannoy S. , Goguel V. , Han K.A. , Isabel G. , Preat T. (2017) **Upregulated energy metabolism in the *Drosophila* mushroom body is the trigger for long-term memory** *Nat Commun* **8**
- Rai M. , Coleman Z. , Curley M. , Nityanandam A. , Platt A. , Robles-Murguía M. , Jiao J. , Finkelstein D. , Wang Y.D. , Xu B. , et al. (2021) **Proteasome stress in skeletal muscle mounts a long-range protective response that delays retinal and brain aging** *Cell Metab* **33**:1137–1154
- Rajan A. , Perrimon N (2012) ***Drosophila* cytokine unpaired 2 regulates physiological homeostasis by remotely controlling insulin secretion** *Cell* **151**:123–137
- Ranganayakulu G. , Elliott D.A. , Harvey R.P. , Olson E.N. (1998) **Divergent roles for NK-2 class homeobox genes in cardiogenesis in flies and mice** *Development* **125**:3037–3048
- Ribeiro C. , Dickson B.J (2010) **Sex peptide receptor and neuronal TOR/S6K signaling modulate nutrient balancing in *Drosophila*** *Curr Biol* **20**:1000–1005
- Saltiel A.R. , Kahn C.R (2001) **Insulin signalling and the regulation of glucose and lipid metabolism** *Nature* **414**:799–806
- Scheiner R. , Steinbach A. , Classen G. , Strudthoff N. , Scholz H (2014) **Octopamine indirectly affects proboscis extension response habituation in *Drosophila melanogaster* by controlling sucrose responsiveness** *J Insect Physiol* **69**:107–117
- Schwaerzel M. , Monastirioti M. , Scholz H. , Friggi-Grelín F. , Birman S. , Heisenberg M (2003) **Dopamine and octopamine differentiate between aversive and appetitive olfactory memories in *Drosophila*** *J Neurosci* **23**:10495–10502
- Semaniuk U. , Strilbytska O. , Malinovska K. , Storey K.B. , Vaiserman A. , Lushchak V. , Lushchak O (2021) **Factors that regulate expression patterns of insulin-like peptides and their association with physiological and metabolic traits in *Drosophila*** *Insect Biochem Mol Biol* **135**

Shingleton A.W. , Das J. , Vinicius L. , Stern D.L (2005) **The temporal requirements for insulin signaling during development in *Drosophila*** *PLoS Biol* **3**

Tennessen J.M. , Barry W.E. , Cox J. , Thummel C.S (2014) **Methods for studying metabolism in *Drosophila*** *Methods* **68**:105–115

Tully T. , Quinn W.G (1985) **Classical conditioning and retention in normal and mutant *Drosophila melanogaster*** *J Comp Physiol A* **157**:263–277

Vargas M.A. , Luo N. , Yamaguchi A. , Kapahi P (2010) **A role for S6 kinase and serotonin in postmating dietary switch and balance of nutrients in *D. melanogaster*** *Curr Biol* **20**:1006–1011

Wigglesworth V.B (1949) **The utilization of reserve substances in *Drosophila* during flight** *J Exp Biol* **26**:150–163

Wu C.L. , Shih M.F. , Lee P.T. , Chiang A.S (2013) **An octopamine-mushroom body circuit modulates the formation of anesthesia-resistant memory in *Drosophila*** *Curr Biol* **23**:2346–2354

Wu Q. , Zhang Y. , Xu J. , Shen P (2005) **Regulation of hunger-driven behaviors by neural ribosomal S6 kinase in *Drosophila*** *Proc Natl Acad Sci U S A* **102**:13289–13294

Yamada T. , Habara O. , Kubo H. , Nishimura T (2018) **Fat body glycogen serves as a metabolic safeguard for the maintenance of sugar levels in *Drosophila*** *Development* **145**

Yamagata N. , Ichinose T. , Aso Y. , Placais P.Y. , Friedrich A.B. , Sima R.J. , Preat T. , Rubin G.M. , Tanimoto H (2015) **Distinct dopamine neurons mediate reward signals for short- and long-term memories** *Proc Natl Acad Sci U S A* **112**:578–583

Youn H. , Kirkhart C. , Chia J. , Scott K (2018) **A subset of octopaminergic neurons that promotes feeding initiation in *Drosophila melanogaster*** *PLoS One* **13**

Zars T. , Fischer M. , Schulz R. , Heisenberg M (2000) **Localization of a short-term memory in *Drosophila*** *Science* **288**:672–675

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Reviewer #1 (Public Review):

The present study by Berger et al. analyzes to what extent memory formation is dependent on available energy reserves. This has been dealt with extensively in the case of aversive memory formation, but only very sparsely in the case of appetitive memory formation. It has long been known that an appetitive memory in flies can only be formed by starvation. However, the authors here additionally show that not only the duration of starvation plays a role, but also determines which form of memory (short- or long-term memory) is formed. The authors demonstrated that internal glycogen stores play a role in this process and that this is achieved through insulin-like signaling in octopaminergic reward neurons that integrates internal energy stores into memory formation. Here, the authors suggest that octopamine plays a role as a negative regulator of different forms of memory.

The study sheds light on an old question, to what extent the octopaminergic neuronal system plays a role in the formation of appetitive memory, since in recent years only the dopaminergic system has been in focus. Furthermore, the data are an interesting contribution to the ongoing debate whether insulin receptors play a role in neurons themselves or in glial cells. The experiments are very well designed and the authors used a variety of behavioural experiments, genetic tools to manipulate neuronal activity and state-of-the-art imaging techniques. In addition, they not only clearly demonstrated that octopamine is a negative regulator of appetitive memory formation, but also proposed a mechanism by which the insulin receptor in octopaminergic neurons senses the internal energy status and then controls the activity of those neurons. The conclusions are mostly supported by the data, but some aspects related to the experimental design, some explanations and literature references need more clarification and revision.

1. Usually, long-term memory (LTM) is tested 24 hours after training. Here, the authors usually refer to LTM as a memory that is tested 6 hours after training. The addition of a

control experiment to show that LTM that the authors observe here lasts longer would increase the power of this study immensely.

2. The authors define here another consolidated memory component as ARM, when they applied a cold-shock 2 hours after training. However, some publications showed that LTM is formed after only one training cycle (Krashes et al 2008, Tempel et al 1983). This makes it difficult to determine, whether appetitive ARM can be formed. Furthermore, one study showed that appetitive ARM is absent after massed training (Colomb et al 2009). Therefore, the conclusion could be also, that different starvation protocols, would lead to different stabilities of LTM. Therefore, additional experiments could help to clarify this opposing explanation. From these results, it can then be concluded either that different stable forms of LTM are formed depending on the starvation state, or that two differently consolidated memory phases (LTM, ARM) are formed, as has already been shown for aversive memory. This is also important for other statements in the manuscript, and therefore the authors should address this. For example, the findings about the insulin receptor (is it two opposing memories or different stabilities of LTM).

Reviewer #2 (Public Review):

How organism physiological state modulates establishment and perdurance of memories is a timely question that the authors aimed at addressing by studying the interplay between energy homeostasis and food-related conditioning in *Drosophila*. Specifically, they studied how starvation modulates the establishment of short-term vs long-term memories and clarified the role of the monoamine Octopamine in food-related conditioning, showing that it is not per se involved in formation of appetitive short-term memories but rather gates memory formation by suppressing LTM when energy levels are high. This work clarifies previously described phenotypes and provides insight about interconnections between energy levels, feeding and formation of short-term and long-term food-related memories. In the absence of population-specific manipulation of octopamine signaling, it however does not reach a circuit-level understanding of how these different processes are integrated.

Strengths

- Previous studies have documented the impact of Octopamine on different aspects of food-related behaviors (regulation of energy homeostasis, feeding, sugar sensing, appetitive memory...), but we currently lack a clear understanding of how these different functions are interconnected. The authors have used a variety of experimental approaches to systematically test the impact of internal energy levels in establishment of appetitive memory and the role of Octopamine in this process.
- The authors have used a range of approaches, performed carefully controlled experiments and produced high quality data.

Weaknesses

- 1- In the *tbh* mutant flies, Tyramine -to- Octopamine conversion is inhibited, resulting not only in a lack of Octopamine, but also in elevated levels of Tyramine. If and how elevated levels of Tyramine contributes to the described phenotypes is unclear. In the current version of the manuscript, only one set of experiments (Figure 2) has been performed using Octopamine agonist. This is particularly important in light of recent published data showing that starvation modifies Tyramine levels.
- 2- Octopamine (and its precursor Tyramine) have been implicated in numerous processes, complicating the analysis of the phenotypes resulting from a general inhibition of *tbh*.

3- The manuscript explores various aspects of the impact of energy levels on food-related behaviors and the underlying sensing and effector mechanism, both in wild-type and *tbh* mutants, making it difficult to follow the flow of the results.

Reviewer #3 (Public Review):

In this manuscript, Berger et al. study how internal energy storage influence learning and memory. Since in *Drosophila melanogaster*, octopamine (OA) is involved in the regulation of energy homeostasis they focus on the roles of OA. To do so they use the tyramine- β -hydroxylase (*Tbh*) mutant that is lacking the neurotransmitter OA and study short term memory (STM), long-term memory (LTM) and anesthesia-resistant memory (ARM). They show that the duration of starvation affects the magnitude of both short- and long-term memory. In addition, they show that OA has a suppressive effect on learning and memory. In terms of energy storage, they show that internal glycogen storage influences how long sucrose is remembered and high glycogen suppresses memory. Finally, they show that insulin-like signaling in octopaminergic neurons, which is also related to internal energy storage, suppresses learning and memory.

This is an important study that extends our knowledge on OA activity in learning and memory and the effects the metabolic state has on learning and memory. The authors nicely use the genetic tools available in flies to try and unravel the complex circuitry of metabolic state level, OA activity and learning and memory.

Nevertheless, I do have some comments that I think require attention:

1. The authors use RNAi to reduce the level of glycogen synthase or glycogen phosphorylase. These manipulations are expected to affect the level of glycogen. Using specific drivers the authors attempt to manipulate glycogen level at the muscles and fat bodies and examine how this affects learning and memory. The conclusions of the authors arise solely from the manipulation intended (i.e. the genetics). However, the authors also directly measured glycogen levels at these organs and those do not follow the manipulation intended, i.e. the RNAi had very limited effect on the glycogen level. Nevertheless, these results are ignored.

2. The authors claim in the summary that OA is not required for STM. However, according to one experiment OA is required for STM as *Tbh* mutants cannot form STM. In another experiment OA is suppressive to STM as wt flies fed with OA cannot form STM. Therefore, it is very difficult to appreciate the actual role of OA on STM.

3. The authors use t-test and ANOVA for most of the statistics, however, they did not perform normality tests. While I am quite sure that most datasets will pass normality test, nevertheless, this is required.

4. While it is logical to assume that OA neurons are upstream to R15A04 DA neurons, I am not sure this really arises from the experiment that is presented here. It is well established that without activity in R15A04 DA neurons there is no LTM. Since OA acts to decrease LTM, can one really conclude anything about the location of OA effect when there is no learning?

5. It is unclear how expression of a dominant negative form of insulin receptor (InR) in OA neurons can rescue the lack of OA due to the *Tbh* mutation. If OA neurons cannot release anything to the presumably downstream DA neurons, how can changing their internal signaling has any effect?

While I stressed some comments that need to be addressed, the overall take-home message of the manuscript is supported and the authors do show that the metabolic state of the

animal affects learning and memory. I do think though, that some more caution is required for some of the conclusions.